Mahmoud Alyosify

AI & Machine Learning Engineer

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SUMMARY

AI and machine learning engineer with an **MSc in Artificial Intelligence from Queen's University, Canada (GPA 4.30 / 4.30)**. I build LLM, RAG and machine learning systems and take them all the way to a working, tested service. Currently an AI Engineer Intern at Injaz Tech; before that, systems engineering on Egypt's national toll network and six years teaching AI to 10,000+ learners.

SKILLS

AI & LLMs: LLMs, RAG, LangChain, LangGraph, fine-tuning (LoRA, QLoRA), prompt engineering, AI agents
Machine learning: PyTorch, TensorFlow, scikit-learn, computer vision, NLP, reinforcement learning
Engineering: Python, SQL, FastAPI, Docker, AWS, Git, Linux, C#, .NET

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo — Oct 2026 – Present

- Built a RAG assistant that answers staff questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt — Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo — Feb 2024 – Mar 2025

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles**.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework** and **SQL Server**.

Online Instructor — Udemy & YouTube (Arabic courses), Remote — 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

MinimalLM — Master's thesis (Python, PyTorch, Hugging Face Transformers) — 2026

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).

HORUS Sentinel — multi-agent intelligence platform (Python, LLM agents, knowledge graph)

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.

SimCLR-Vision-SSL — self-supervised image model (PyTorch, ONNX, FAISS) — 2026

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada — Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt — Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

IBM AI Developer Professional Certificate (IBM, 2026) · Deep Learning Specialization (DeepLearning.AI, 2026) · Machine Learning Specialization (DeepLearning.AI) · AWS Academy: ML Foundations, ML for NLP, Data Engineering · 1st Place, Smart Cities Hackathon 2022 (77 university teams) · Semi-Finalist, Microsoft Imagine Cup 2023

LANGUAGES

Arabic (native) · English (professional working proficiency)